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Master in Space Engineering Spacecraft Attitude Dynamics course

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	Assigned specification	Modifications	Motivation for modification
Platform	≥ 500 kg satellite		
Attitude parameters	Quaternions		
Mandatory sensor	Earth sensor	Added additional sun sensor	Unobservability in yaw angle
Actuators	Two RWs, one IW		

Contents

1	Introduction	1
1.1	Mission overview and requirements	1
2	Description of the model	2
2.1	Spacecraft characteristics	2
2.2	Dynamics and Kinematics	3
2.3	Environment	4
2.3.1	Gravity gradient torque	4
2.3.2	Solar radiation pressure torque	5
2.3.3	Magnetic field torque	5
2.3.4	Air drag torque	6
2.3.5	Torque comparison	6
2.3.6	Eclipse	6
2.4	Sensors	7
2.4.1	Horizon sensor	8
2.4.2	Sun sensor	8
2.5	Actuators	8
2.5.1	Software	9
2.5.2	Hardware	9
2.5.3	Sizing of reaction wheels	9
3	Attitude Determination and Control System	11
3.1	Wahba's problem: q-method	11
3.2	Kalman Observer	12
3.3	De-tumbling	14
3.4	Slew manoeuvre	14
3.5	Pointing	15
3.6	Performance evaluation	15
4	Results	16
5	Discussion	19

Symbology

Symbol	Meaning	Unit
J	Inertia Matrix	$kg \cdot m^2$
I_x	Inertia momentum along x axis	$kg \cdot m^2$
I_y	Inertia momentum along y axis	$kg \cdot m^2$
I_z	Inertia momentum along z axis	$kg \cdot m^2$
ω	Angular velocity	rad/s
T	Total external torque	Nm
T_{eff}	Effective applied torque	Nm
M_d	External disturbances	Nm
q	Quaternion	—
e	Euler rotation axis	—
ϕ	Euler rotation angle	rad
DCM	Direction Cosine Matrix	—
T_{GG}	Gravity gradient torque	Nm
T_{SRP}	Solar radiation pressure torque	Nm
T_{MF}	Magnetic field torque	Nm
$\hat{N}$	Normal to surface	—
$\hat{S}$	Sun direction	—
ρ_s	Specular reflection coefficient	—
ρ_d	Diffuse reflection coefficient	—
$R_{\oplus}$	Earth radius	km
R_S	Sun radius	km
h_r	Reaction wheel angular momentum vector	$kg \cdot m^2/s$
h_{IW}	Inertia wheel angular momentum	$kg \cdot m^2/s$
M_c	Commanded control torque	Nm
K_P	Proportional gain for control	—
K_D	Derivative gain for control	—

Table 1: Symbology.

1 Introduction

The aim of this project is to develop and demonstrate effective control techniques for a GEO communication satellite. In this report, the mission requirements are stated, the models for the actuators and sensors are presented, and the design of the Attitude Determination and Control System (ADCS) is analysed. Active attitude determination, de-tumbling, slew manoeuvres and pointing phases are considered in the study. Figure 1 represents a block scheme of the ADCS architecture, including the different models and algorithms created.

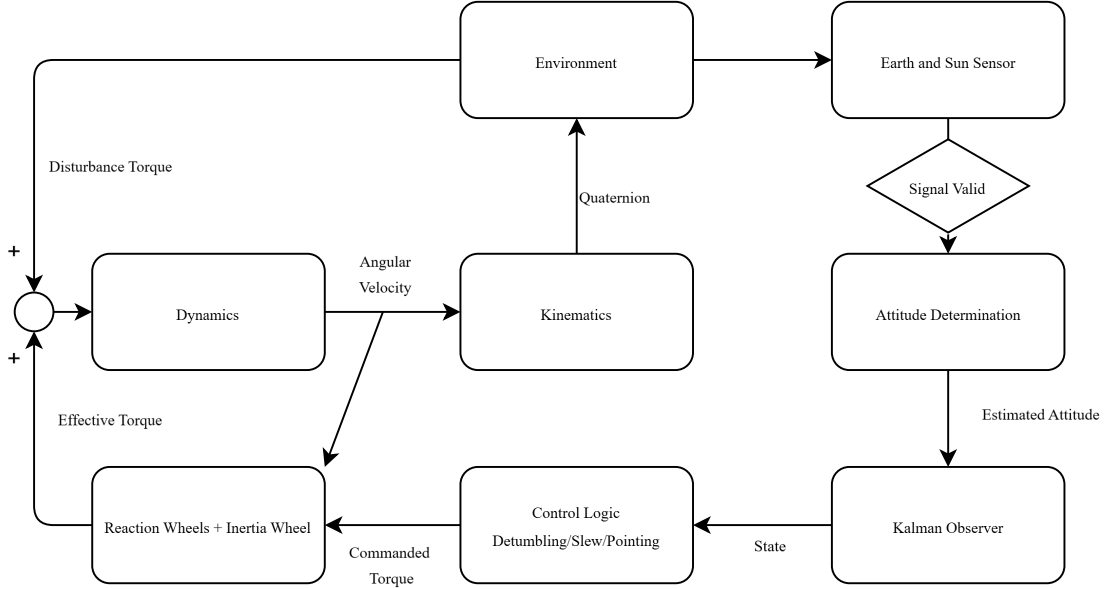


Figure 1: ADCS architecture scheme.

1.1 Mission overview and requirements

The assigned satellite's characteristics imposed a platform of more than 500 kg (large satellite), with at least a horizon sensor, an inertia wheels and two reaction wheels. To enhance the realism of the project, the final satellite's characteristics have been modelled based on the GOES-R 16, part of the Geostationary Operational Environmental Satellite series [5]. It is a weather satellite that operates in geostationary orbit (GEO). The choice of this satellite as a reference is strategic, not only because its dimensions match the project specifications but also because of the mission for which it was designed. GOES-R 16 is a geostationary satellite that must maintain a continuous orientation pointing to the Earth. This requirement is perfectly in line with the sensor configuration imposed in the project, making it an ideal model for the study. Aligned with the GOES-R 16 satellite, the nominal orbit considered for this study is geostationary (GEO), characterized by a semimajor axis of $a = 42164$ km, with zero inclination and eccentricity. This orbit ensures that the satellite remains stationary relative to a fixed point on Earth, a crucial feature for consistent Earth observation. The final objective of the mission is to actively track the Earth for nadir-pointing, maintaining a continuous orientation towards the planet's surface. This mission goal is achieved through the following operational phases:

- **de-tumbling:** this phase occurs immediately after the satellite's deployment into orbit, where the satellite's angular velocity is stabilized. Residual rotational motion from the launch must

be dampened to ensure the spacecraft is in a controlled state, enabling subsequent precision maneuvers. The mission requires the residual velocity to be less than 10^{-5} rad/s.

- **Slew Manoeuvrer:** once de-tumbling is complete, the satellite performs a slew manoeuvrer to orient itself correctly. This involves rotating the spacecraft to align its primary observation instruments or antennas with the Earth, setting up the initial nadir-pointing configuration.
- **Target Tracking:** in this phase, the satellite continuously adjusts its attitude to maintain Earth-pointing orientation. Using onboard sensors and actuators, it compensates for perturbations, ensuring accurate and consistent alignment with the nadir direction. The pointing requirement implies an angular deviation from the nominal nadir pointing of no more than 1° during one entire orbital period. Power generation requirements also impose a yaw angle requirement to be less than 10° .

It is important to note that during both de-tumbling and slew manoeuvres, the solar panel remains stowed, only to be deployed later during the pointing phase.

2 Description of the model

2.1 Spacecraft characteristics

The satellite model has a total mass of 1120 kg including a solar panel, and it is studied in two different configurations: with the solar panel stowed and deployed. The properties of the satellite are listed in Table 2. Furthermore, the spacecraft is equipped with a sun sensor in addition to the mandatory horizon sensor, and three wheels aligned with the principal axis (two reaction wheels and one inertia wheel).

	Satellite core	Solar panel
Mass [kg]	1000	120
Dimensions [m]	1.5x1.5x1	3x1
	Solar panel stowed	Solar panel deployed
Inertia tensor [$\text{kg} \cdot \text{m}^2$]	$\begin{bmatrix} 357.63 & 0 & 0 \\ 0 & 357.63 & 0 \\ 0 & 0 & 471.70 \end{bmatrix}$	$\begin{bmatrix} 968.33 & 0 & 0 \\ 0 & 978.33 & 0 \\ 0 & 0 & 992.50 \end{bmatrix}$

Table 2: Properties of the satellite.

2.2 Dynamics and Kinematics

The satellite is modelled as a rigid body, and its rotational dynamics are governed by Euler's equations:

$$\mathbf{J}\dot{\boldsymbol{\omega}} = (\mathbf{J}\boldsymbol{\omega}) \times \boldsymbol{\omega} + \mathbf{T} \quad (1)$$

where:

- $\mathbf{J}$: inertia matrix in body frame.
- $\boldsymbol{\omega}$: angular velocities.
- $\mathbf{T}$: total external torque, sum of the effective applied torque and external disturbances ($\mathbf{T} = \mathbf{T}_{eff} + \mathbf{M}_d$).

By integrating these equations, the angular velocity $\boldsymbol{\omega}$ is determined, which is then used as an input for the kinematic analysis to compute the spacecraft's orientation.

For attitude representation, quaternions were assigned as parametrization for the project. Quaternions are particularly advantageous due to their ability to avoid singularities inherent in other methods like Euler angles. They can be defined as:

$$\begin{cases} q_1 &= e_1 \sin\left(\frac{\phi}{2}\right) \\ q_2 &= e_2 \sin\left(\frac{\phi}{2}\right) \\ q_3 &= e_3 \sin\left(\frac{\phi}{2}\right) \\ q_4 &= \cos\left(\frac{\phi}{2}\right) \end{cases} \quad (2)$$

where e_1 , e_2 and e_3 are the components of the rotation axis, and ϕ is the rotation angle. Quaternions satisfy the normalization condition:

$$q_1^2 + q_2^2 + q_3^2 + q_4^2 = 1 \quad (3)$$

And the kinematics are governed by:

$$\dot{q}(t) = \frac{1}{2}\Omega(t)q(t) \quad (4)$$

with:

$$\Omega(t) = \begin{bmatrix} 0 & \omega_z & -\omega_y & \omega_x \\ -\omega_z & 0 & \omega_x & \omega_y \\ \omega_y & -\omega_x & 0 & \omega_z \\ -\omega_x & -\omega_y & -\omega_z & 0 \end{bmatrix} \quad (5)$$

While quaternions eliminate the gimbal lock issues of Euler angles, they introduce mathematical ambiguities, as both q and $-q$ represent the same rotation. This ambiguity can introduce inconsistencies when quaternions are used in attitude control or state estimation. To resolve this, the system ensures a consistent representation by enforcing the following specific condition:

$$\text{If } q_4 < 0, \quad q \rightarrow -q \quad (6)$$

By enforcing $q_4 > 0$, the system eliminates ambiguities and ensures stability even if the quaternion flips unexpectedly.

2.3 Environment

The satellite, during its orbital motion around the Earth, is constantly influenced by different disturbances, whose magnitude is strongly affected by the physical characteristics of the satellite (such as weight and shape), as well as by the characteristics of its orbit. Such disturbances generate torques that tend to change the attitude of the satellite (as well as the shape of its orbit). The main disturbances that can be encountered are:

- Gravity gradient (T_{GG}).
- Solar radiation pressure (T_{SRP}).
- Air drag (T_{AD}).
- Magnetic field (T_{MF}).

In addition to this, dealing with a geostationary orbit, the satellite will experience periods of darkness caused by solar eclipses. This aspect will also be considered in the satellite's motion, as it will affect the effect of solar radiation pressure and the functioning of the sun sensors.

2.3.1 Gravity gradient torque

The gravity gradient torque is caused by the interaction between the satellite and the Earth's gravitational field, which is not uniform. The contribution of the gravity gradient torque can be modelled by noting that it strongly depends on the orientation of the satellite's principal axes with respect to the LV-LH frame, as well as on the three principal moments of inertia. Its contribution is introduced into the right hand side of the Euler equations in this way:

$$T_{GG} = -3n^2(J\mathbf{c} \times \mathbf{c}) \quad (7)$$

where the vector $\mathbf{c}$ is defined as:

$$\mathbf{c} = A_{B/L} \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} \quad (8)$$

$A_{B/L}$ is the DCM which describes the attitude of the satellite with respect to the LVLH frame. Moreover n is the angular velocity of LVLH frame and J is the inertia matrix.

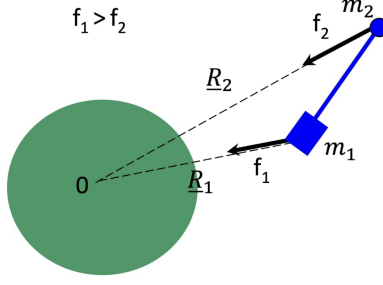


Figure 2: Gravity gradient action.

2.3.2 Solar radiation pressure torque

The solar radiation pressure torque is caused by the photons expelled by the Sun that impact the surface of the satellite, exchanging momentum with it in the process. All the surfaces of the satellite are subjected to this action, and the model that describes the total moment caused by SRP is as follows:

$$T_{SRP} = \begin{cases} \sum_{i=1}^n \mathbf{r}_i \times \mathbf{F}_i & \hat{S}_B \cdot \hat{N}_{Bi} > 0 \\ 0 & \hat{S}_B \cdot \hat{N}_{Bi} \leq 0 \end{cases} \quad (9)$$

with: $\mathbf{F}_i = -PA_i(\hat{S}_B \cdot \hat{N}_{Bi}) \left[(1 - \rho_s)\hat{S}_B + (2\rho_s(\hat{S}_B \cdot \hat{N}_{Bi}) + \frac{2}{3}\rho_d)\hat{N}_{Bi} \right]$, and where:

- $\hat{N}_{Bi}$: normals to each surface of the spacecraft, so 6 vectors for the main body and 2 for both sides of the panel.
- $\hat{S}_B$: Sun direction expressed in body frame.
- P : average radiation pressure $\rightarrow P = \frac{F_e}{c}$, where $F_e = 1358 \text{ W/m}^2$ and $c = 3 \cdot 10^8 \text{ m/s}$.
- ρ_s : specular reflection coefficient (S/C Body: 0.5; S/C Panel: 0.1).
- ρ_d : diffuse reflection coefficient (S/C Body: 0.1; S/C Panel: 0.1).

2.3.3 Magnetic field torque

The magnetic torque is caused by the interaction between the Earth magnetic field and the local magnetic field of the satellite, generated mainly by the electronic equipment. Since the satellite under consideration is in geostationary orbit, the Earth's magnetic field can be approximated as a dipole, simplifying the modelling of the generated torque, which is as follows:

$$T_{MF} = \mathbf{j}_B \times \mathbf{b}_B \quad (10)$$

where $\mathbf{b}_B$ is the magnetic field of earth in the body frame computed as:

$$\mathbf{b}_B = A_{B/N} \frac{R_\oplus H_0}{r^3} [3(\hat{m} \cdot \hat{r})\hat{r} - \hat{m}] \quad (11)$$

and where $\mathbf{j}_B$ has been selected with a conservative value of $[0.1 \ 0.1 \ 0.1]^T$. Additionally, it is important to note that the Earth's magnetic field has been modelled on the basis of the International Geomagnetic Reference Field (IGRF) model and H_0 is built as:

$$H_0 = \sqrt{(g_1^0)^2 + (g_1^1)^2 + (h_1^1)^2} \quad (12)$$

where g_1^0, g_1^1, h_1^1 are tabulated parameters.

2.3.4 Air drag torque

In general the air drag torque is generated from the interaction between the atmosphere and the satellite and derives from the action of aerodynamic forces. Air drag torque has high values for orbits with an altitude below 400 km, while for orbits with very high altitude, or even geostationary ones, it is negligible because the Earth's atmosphere extends up to approximately 1000 km. Due to this, the air drag torque has not been considered in the analysis from the beginning and consequently has not been modelled.

2.3.5 Torque comparison

In order to decide which disturbance torques to include in the model, the maximum torques were calculated using the simplified models for each of them. The resulting torques are shown in Table 3.

Torque	Panel stowed $[Nm]$	Panel deployed $[Nm]$
T_{GG}	9.0983×10^{-7}	1.9276×10^{-7}
T_{MF}	3.5623×10^{-8}	3.5623×10^{-8}
T_{SRP}	1.5278×10^{-5}	4.0740×10^{-5}

Table 3: Maximum expected disturbance torques.

According to the results, the two maximum disturbances, solar radiation pressure and gravity gradient, were selected. Furthermore, it can be observed that such a small moment owing to the magnetic field was to be expected, given the large distance from Earth that results from the satellite being in a geostationary orbit. The growth in the maximum moment in the third phase (panel deployed), associated with SRP, is attributed to the satellite's asymmetry caused by the presence of a single large solar panel.

2.3.6 Eclipse

During its motion, the satellite experiences periods of shadow caused by solar eclipses. This phenomenon must be taken into account in the simulation of its trajectory, and a conical model has been followed to represent it [7]. The following quantities are defined:

- $\mathbf{r}_B = -\mathbf{r}_{S/C-P}$: vector from satellite to Earth.
- $\mathbf{r}_S = -\mathbf{r}_{S/C-P} + \mathbf{r}_{Sun-P}$: vector from satellite to Sun.

- $H_B = \arcsin\left(\frac{R_\oplus}{|\mathbf{r}_B|}\right)$: apparent Earth's radius as seen by the satellite.
- $H_S = \arcsin\left(\frac{R_S}{|\mathbf{r}_S|}\right)$: apparent Sun radius as seen by satellite.
- $\gamma = \arccos\left(\frac{\mathbf{r}_B \cdot \mathbf{r}_S}{|\mathbf{r}_B||\mathbf{r}_S|}\right)$: angle separation between Earth and Sun as seen by the satellite.

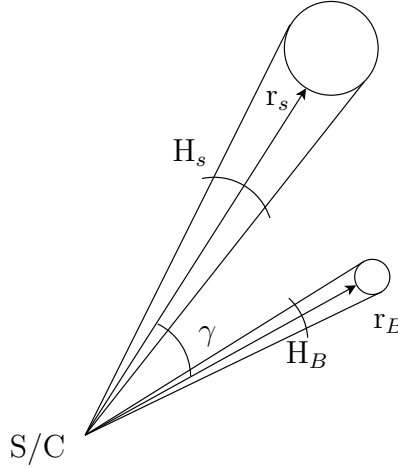


Figure 3: Eclipse conic model representation.

$R_\oplus$ and R_S are the radii of the Earth and Sun. Once the definitions of H_B , H_S and γ are given, and the geometric representation of the eclipse problem is introduced in Figure 3, it becomes clear that the satellite is not in eclipse as long as:

$$\gamma > H_B + H_S \quad (13)$$

Consequently, a logic based on this has been implemented to regulate the effect of the solar radiation pressure torque on the satellite and to control the operation of the sun sensor.

2.4 Sensors

As introduced before, the horizon sensor is assigned as mandatory for the project. Even if the single earth sensor could be sufficient to determine the full state of the spacecraft, by using an observer, a sun sensor is added to increase the precision of the attitude propagation and keeping the complete state updated, enhancing accuracy and reliability in attitude determination. Being in GEO, sun sensors are particularly advantageous because eclipses occur for a shorter amount of time than in LEO. Magnetometers were not considered as they are not used in GEO.

2.4.1 Horizon sensor

As a reference for the horizon sensor, the Infrared Earth Sensor (IRES), manufactured by Leonardo, has been considered [3]. This is a two-axis infrared Earth horizon sensor for accurate measurement of pitch and roll attitude angles. It has been chosen because it is specifically designed for operation in GEO, particularly on spacecraft with three-axis stabilization systems.

The model has four different operational modes, but only two of them are considered in the project. The characteristics are shown in Table 4. The modes considered are the following:

- **Pointing mode linear range (PM):** provides a narrow scanning, used for the final phase of nadir-pointing.
- **Chord mode linear range (CM):** single beam crossing that provides a narrow scan, used in the de-tumbling and slew manoeuvre.

Measurement range	Output data rate	Accuracy	Bias error
PM: $\pm 5.5^\circ$ pitch; $\pm 2.5^\circ$ roll	10 Hz	$< 0.05^\circ (3\sigma)$	$< 0.02^\circ$
CM: $\pm 23^\circ$ pitch; $\pm 14^\circ$ roll			

Table 4: Specifications of the IRES horizon sensor [3].

2.4.2 Sun sensor

A sun sensor determines the orientation of the spacecraft with respect to the Sun by measuring the angle of incoming sunlight. For this project, the Leonardo Smart Sun Sensor (S3) has been selected as a reference [4]. The S3 is a two-axis solar sensor based on an Active Pixel Sensor (APS) detector. It has been optimized for the commercial market of Earth Observation satellites and, thanks to the intrinsic radiation hardness of the APS detector, it is also suitable for GEO telecommunication spacecraft.

To ensure greater visibility of the Sun under varying spacecraft orientations, two sun sensors, which have the z-axis direction of their reference frame aligned with the y and -y directions of the body frame, have been incorporated into the design, both with resized $\pm 70^\circ$ FOV. This redundancy helps maintain the accuracy of attitude determination. The key characteristics of the S3 are detailed in Table 5.

Field of view (FOV)	Output data rate	Accuracy	Bias error
$\pm 70^\circ$	10 Hz	$< 0.02^\circ (2\sigma)$	$< 0.01^\circ$

Table 5: Specifications of the S3 sun sensor [4].

2.5 Actuators

Onboard the satellite are two reaction wheels and one inertia wheel. All three are of the same Honeywell HR-16 100 Nms variant [2]. Two reaction wheels are operating at zero angular velocity

aligned with the x and y axes of body frame. Depending on the mission phase, the third reaction wheel aligned with the spacecraft z-direction will be operating either around zero or a mode as an inertial wheel with a nominal relative angular momentum. It should be noted that this configuration does not provide any redundancy and would therefore not be used in practise as the main actuators for attitude control.

HR-16 100 Nms	
Reaction torque	0.1 Nm
Momentum	100 Nms
Wheel speed	± 6000 rpm
Mass	12 kg
Moment of inertia	$0.1592 \text{ kg} \cdot \text{m}^2$

Table 6: Specifications of the HR-16 reaction wheels [2].

2.5.1 Software

Summarizing the effect of the reaction wheels on the spacecraft dynamics as an internally applied torque and solving for the required reaction torque yields the general control problem as follows :

$$\dot{\mathbf{h}}_r = A^*(-\mathbf{M}_c - \hat{\boldsymbol{\omega}} \times A\mathbf{h}_r - \hat{\boldsymbol{\omega}} \times (J\hat{\boldsymbol{\omega}})) \quad (14)$$

where $\mathbf{M}_c$ is the torque commanded by the control, A is the 3×3 matrix of direction cosines giving the spin vector directions for each actuator and $\hat{\boldsymbol{\omega}}$ is the estimated angular velocity of the spacecraft. Since the reaction wheels are aligned with the main axis of the spacecraft, $A = I_{3 \times 3}$. Multiplication with the vector $\mathbf{h}_r$ whose elements are the scalar values of the angular momentum of each reaction wheel. Solving for the reaction torque in this way compensates for the gyroscopic effect of the reaction wheels having a non-zero spin when an estimate of the spacecraft's angular velocity is obtained.

2.5.2 Hardware

Once the required torque $\mathbf{M}_c$ is computed, the actuators hardware supplies the spacecraft with an actual torque which is computed as follows :

$$\mathbf{T}_{eff} = -\boldsymbol{\omega} \times A\mathbf{h}_r - A\dot{\mathbf{h}}_r \quad (15)$$

Where $\boldsymbol{\omega}$ is the real angular velocity of the spacecraft.

2.5.3 Sizing of reaction wheels

There are four aspects to take into account for the sizing of the actuators:

- The torque needed to effectively de-tumble the spacecraft.
- The torque needed to counteract the environmental disturbances during the manoeuvres.

- The maximum momentum the actuators can absorb to de-tumble the spacecraft without saturating.
- The maximum momentum reached when counteracting the disturbing torques.

Due to the small field of view of the Earth sensor, the time window in which a final state at rest is guaranteed during the de-tumbling imposes a minimum reaction torque. During nominal pointing, the orbit average of the disturbing torques will be close to zero. This is due to the fact that the SRP torque is cyclical and that the principal inertial axis is aligned with the gravity gradient resulting in an almost zero change in angular momentum over the full orbital period. This can be seen in figure 4.

The reaction wheels must be able to absorb the initial angular momentum of the spacecraft at the time of release from the launcher. For the spacecraft to come to rest, the total angular momentum must be zero, which in turn means that $A\mathbf{h}_r = -\mathbf{h}_{SC}^B = -A_{B/N}\mathbf{h}_{SC}^N$. When $A_{B/N}\mathbf{h}_{SC}^N$ aligns with one of the spin axes of the reaction wheels, then the momentum of the corresponding wheel will be highest. This means that the maximum relative angular momentum must be greater than $|A_{B/N}\mathbf{h}_{SC}^N| = |\mathbf{h}_{SC}^N| = |J_{stowed}\boldsymbol{\omega}_{0MAX}| = 6.98 \text{ Nms}$.

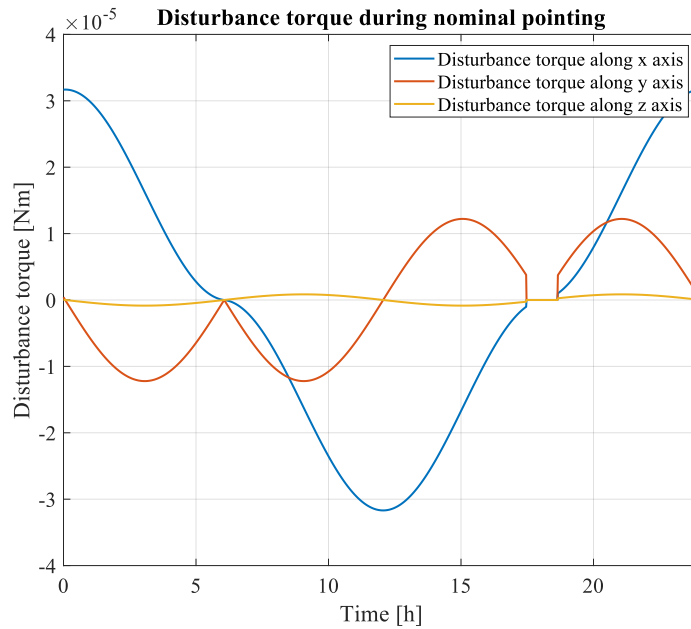


Figure 4: Effect of gravity gradient and SRP torque on the spacecraft during nominal pointing.

Accounting for the worst case disturbing torque, $T_{Dmax} = \max(T_{GG}, T_{SRP}, T_{MF}) = 4.074 \times 10^{-5} \text{ Nms}$ applied constantly over half the duration of the orbit. The requirement of keeping two-axis pointing means an internal change of angular momentum equal to 1.76 Nms . Thus including a suitable margin and also taking into account the 70 Nms nominal momentum of the inertia wheel means that the 100 Nms variant can be chosen. For simplicity all actuators are chosen to be of the same variant.

3 Attitude Determination and Control System

The attitude control system is tasked with maintaining or adjusting the spacecraft's orientation to achieve the desired pointing accuracy. This system works in tandem with the attitude determination process, which computes the spacecraft orientation using quaternion-based solutions. The determination process is performed through Wahba's problem solution and Kalman observer.

Using the attitude information, the control system applies the necessary torques through the wheels to stabilize the spacecraft or perform the expected manoeuvres. The control strategy ensures stability and efficiency, even under the influence of external disturbances or sensor uncertainties.

3.1 Wahba's problem: q-method

The knowledge of two directions in body frame, provided by the Earth sensor and the Sun sensor (when both are operational), and the knowledge of the same directions in the LV-LH frame allows the determination of the satellite's attitude using specific algorithms. In this case, the algorithm associated with the Wahba's problem has been employed, where the goal is to minimize a weighted error function to achieve an attitude as close as possible to the actual one. In this context, the function to be minimized is the following:

$$J_w = \frac{1}{2} \sum_{i=1}^N \alpha_i |\mathbf{s}_{Bi} - A_{B/L} \mathbf{v}_{Li}|^2 \quad (16)$$

where:

- N : number of directions known from the sensors.
- α_i : weights for the reliability of the directions. In our case $\mathbf{s}_1 = \mathbf{p}$ comes from the Sun sensor, $\mathbf{q}$ comes from Earth sensor, $\mathbf{s}_2 = \frac{\mathbf{p} \times \mathbf{q}}{|\mathbf{p} \times \mathbf{q}|}$ and $\mathbf{s}_3 = \mathbf{p} \times \mathbf{s}_2$ are combinations of the Earth and Sun directions. Therefore, the following coefficients were computed: $\alpha_1 = 0.5$, $\alpha_2 = 0.2525$, $\alpha_3 = 0.2475$.
- $\mathbf{s}_{Bi}$ are the direction vectors in body frame.
- $\mathbf{v}_{Li}$ are the direction vectors in LVLH frame.

Revisiting Wahba's problem and considering quaternions as the kinematic model, it is possible to arrive at a new formulation, the q-method. The weighted error function, in this case to be maximized, is:

$$\tilde{J}_w = q^T K q, \quad \text{with the constraint} \quad q^T q = 1 \quad (17)$$

where the following quantities are introduced:

- $K = \begin{bmatrix} S - \sigma I & z \\ z^T & \sigma \end{bmatrix}$

- $\sigma = \text{tr}(B)$
- $B = \sum_{i=1}^N \alpha_i s_{Bi} v_{Ni}^T$
- $S = B + B^T$
- $Z = [B_{23} - B_{32}, B_{31} - B_{13}, B_{12} - B_{21}]$

In the solution of Wahba's problem, it follows that the maximum eigenvalue of K maximizes the cost function, and the eigenvector associated with it corresponds to the quaternion that determines the attitude of the satellite.

3.2 Kalman Observer

Due to the selection of sensors at dispose, it is impossible to have the attitude determination working all the time. This is either caused by the two measured direction being almost parallel in some positions in the orbit or by the absence of a sensor output. The latter happens specifically during the eclipse phase. Furthermore no gyroscopes are included to retrieve the angular velocities. For these reasons, an observer was implemented to both propagate the state during the absence of the sun sensor and to filter the Wahba's solution and reconstruct the angular velocity when both sensors are reliable. The chosen state is the following :

$$\mathbf{x} = [\dot{\boldsymbol{\alpha}}; \boldsymbol{\alpha}] \quad (18)$$

where $\boldsymbol{\alpha}$ represents the angles with respect to the local vertical local horizontal frame and $\dot{\boldsymbol{\alpha}}$ represents the angular velocities relative to the LV-LH frame. The linearized model around the nominal pointing was chosen. Further steady-state hypothesis was made to solve the Riccati equation considering the gain matrix P to be constant in time. Thus, the problem was simplified into solving an algebraic matrix equation:

$$0 = AP + PA^T + FWF^T - PC^TV^{-1}CP \quad (19)$$

where A is the system dynamics matrix and had to be calculated differently for inertia wheel not active and active scenarios. For inertia wheel not nominally working it was derived to be:

$$A = \begin{pmatrix} 0 & -n \frac{(I_z - I_y - I_x)}{I_x} & 0 & -n^2 \frac{(I_z - I_y)}{I_x} & 0 & 0 \\ -n \frac{(I_x + I_y - I_z)}{I_y} & 0 & 0 & 0 & -n^2 \frac{(I_z - I_x)}{I_y} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \end{pmatrix} \quad (20)$$

while for the case with an active inertia wheel the dynamics matrix was computed as follows:

$$A_{IW} = \begin{pmatrix} 0 & -n\frac{(I_z-I_y-I_x)}{I_x} & 0 & -n^2\frac{(I_z-I_y)}{I_x} - n\frac{h_{IW}}{I_x} & 0 & 0 \\ -n\frac{(I_x+I_y-I_z)}{I_y} + n\frac{h_{IW}}{I_y} & 0 & 0 & 0 & -n^2\frac{(I_z-I_x)}{I_y} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 \end{pmatrix} \quad (21)$$

In both scenarios, two other subcases are to be determined in order to define output matrix C: full state available (both sensors active), partial state available (only earth sensor active).

$$C_{full\ state} = \begin{pmatrix} 1 & 0 & 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 & 1 & 0 \\ 0 & 0 & 1 & 0 & 0 & 1 \end{pmatrix}; \quad C_{partial\ state} = \begin{pmatrix} 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix} \quad (22)$$

The left-hand side identity matrix in the $C_{full\ state}$ matrix helps to have a faster estimation of the angular velocities. This helps in the de-tumbling and slew manoeuvre phases. However, in the pointing it is avoided in order to compensate for the increasing errors during indeterminate attitude phases. The white noise associated with the external disturbances W was found with the variance of the external disturbances. Following this choice, F matrix is equal to the control input matrix B:

$$B = \left(J^{-1} \left| \begin{array}{ccc} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{array} \right. \right) \quad (23)$$

Finally, the white noise matrix V was computed by calculating the variance of the white noise from the sensors. At this point, the state can be computed by integrating the following:

$$\dot{\mathbf{x}} = \hat{A}\mathbf{x} + B\mathbf{u} + L\mathbf{y} \quad (24)$$

where $\mathbf{u}$ is the torque commanded by the control logic and $\mathbf{y}$ is the input state coming from the sensors/attitude determination. The input is the complete angles with respect to LV-LH or only α_x , α_y when only the Earth sensor is available. The matrices $L, \hat{A}$ are defined as follows:

$$L = V^{-1}CP; \quad \hat{A} = A - LC \quad (25)$$

Albeit the linearization around the nominal pointing, when all sensors are active, the Kalman filter outputs a reliable state, from which the angular velocity with respect to the inertial frame can be easily reconstructed. This is mainly important due to the fact that the angular velocity being only estimated through the Kalman observer. This angular velocity is then used for the de-tumbling control. While the state is directly representative of the error with reference to the LV-LH frame, both in angles and in angular velocities. Thus, by transforming α into the quaternion, it can be directly used for the implementation of the slew and pointing control.

3.3 De-tumbling

The de-tumbling phase directly follows the injection of the spacecraft into orbit. The manoeuvre is accomplished bringing the spacecraft to a manageable angular velocity. The controller receives the estimate for the angular velocity through the Kalman observer and commands a torque from the actuators proportional to the angular velocity vector in order to reduce the spacecraft angular velocity. If only the Earth sensor is available, then the propagated state from the observer is used for the control. When no signal is acquired, the control action is stopped. In this situation, the spacecraft will continue rotating until the Earth comes into the sensor's field of view. From there, the control is resumed.

$$M_c = K\omega \quad (26)$$

Observation on the performances showed that there is no need for a derivative part, and thus its absence is justified. Regarding the choice of the control parameter K , associated with the angular velocity, its value was calculated based on plausible values of the natural frequency w_n for a closed-loop system given a large satellite in geostationary orbit. Specifically, according to the literature, suitable values of natural frequency are around 0.1, providing a reasonable response time and an acceptable overshoot. Starting from this value, K was calculated using the following expression:

$$K = 2\xi w_n I_z \quad (27)$$

Subsequently, its value was slightly adjusted by analysing the closed-loop system stability, response time and verifying that the actuators were always capable of satisfying the required torque. After this process, the following values were chosen for the de-tumbling phase: $K_x = 60.038$, $K_y = 60.038$, $K_z = 60.038$.

3.4 Slew manoeuvre

Once the spacecraft is brought to a inertially non-rotating state, a slew manoeuvre is carried out to adjust the attitude from its current attitude to the nominal nadir-pointing one. This is conveyed by the implemented non-linear Lyapunov control law. Guaranteeing both global asymptotic stability [1] and control performance in terms of overshoot, rise time and settling time, the following control potential was chosen:

$$H(q_{e4}) = 1 - q_{e4}^2, \quad (28)$$

yielding the following expression for the commanded torque:

$$M_c = 2K_P \frac{\partial H(q_{e4})}{\partial q_{e4}} \mathbf{q}_e + K_D \dot{\mathbf{\alpha}} = 2K_P \mathbf{q}_e q_{e4} + K_D \dot{\mathbf{\alpha}}, \quad (29)$$

When the error is small, the control becomes linear due to the fact that $q_{e4} \approx 1$ meaning standard linear PD tuning methods can be employed, substituting $\bar{K}_P = 2 \cdot K_P$ and $\bar{K}_D = K_D$.

The inputs needed by the control are once again the $\dot{\mathbf{\alpha}}$ from the state vector, that represent the angular velocities with respect to the LV-LH frame, which means that they are directly the angular velocity error needed. The error quaternion with respect to the LV-LH frame is obtained through the

state α . In order to have a control that follows an optimal transition to the new attitude, the inputs are filtered with a first-order low-pass filter. So, the commanded torque can be computed as:

Regarding the control parameters for the slew manoeuvre, the value of K_D , associated with the angular velocity, was retained as calculated in the de-tumbling phase. This value, derived from plausible natural frequency values for a large satellite in geostationary orbit, ensures a reasonable response time and adequate stability. For K_D , associated with the quaternion error, its value was calculated using the following formula:

$$K_P = w_n^2 I_z \quad (30)$$

This formula links the natural frequency with the moment of inertia I_z , ensuring a balanced response during the attitude adjustment manoeuvre. After these initial calculations, both parameters were slightly adjusted based on the closed-loop system stability analysis and response time requirements. The final chosen values for the slew phase are as follows: $\bar{K}_{Dx} = 66.038$, $\bar{K}_{Dy} = 66.038$, $\bar{K}_{Dz} = 66.038$, $\bar{K}_{Px} = 4.717$, $\bar{K}_{Py} = 4.717$, $\bar{K}_{Pz} = 4.717$.

3.5 Pointing

Once the new nadir-pointing attitude is achieved, it must be possible to satisfy the requirements for the pointing, in terms of angles with respect to the LV-LH, during the complete orbit coasting. This translates into a control during periods also without sun sensor input. Pointing is achieved through passive and active control.

In this phase, the inertia wheel is rotating at its nominal velocity around the z-axis, providing passive stability around the pitch and roll axis. The presence of external torque actuators is assumed meaning the inertia wheel can be spun up in between the slew- and pointing phase without changing the spacecraft's total angular momentum. Furthermore, the same control logic as used in the slew manoeuvre is applied with reduced values of K_P and K_D by a factor of 15 to take into account the significantly lower control needed in near-static conditions. This allows for a more energy-efficient response while maintaining system stability and precision.

3.6 Performance evaluation

Each phase was modelled in a different *Simulink* model, but the initial conditions for the subsequent models are taken from the final conditions of the previous model. These include angular velocity, attitude, simulation time, and true anomaly, and allow to study a complete mission. For clarity, the initial position in orbit and the FOV of the sensor can be appreciated in Figure 5.

The data on orbit insertion true anomaly, maximum release velocities inspired by the launcher provided ones [6], and maximum initial angles with respect to the LVLH frame can be found in Table 7.

Constrained random values were used to run different simulations and gather data. After a time period of 1 hour from the mission beginning, the slew manoeuvre is started and run for another hour. Finally, the main pointing performance across a complete orbit is compared to the performance with only the inertia wheel active. Without any control, because of having a single solar panel, the SRP is

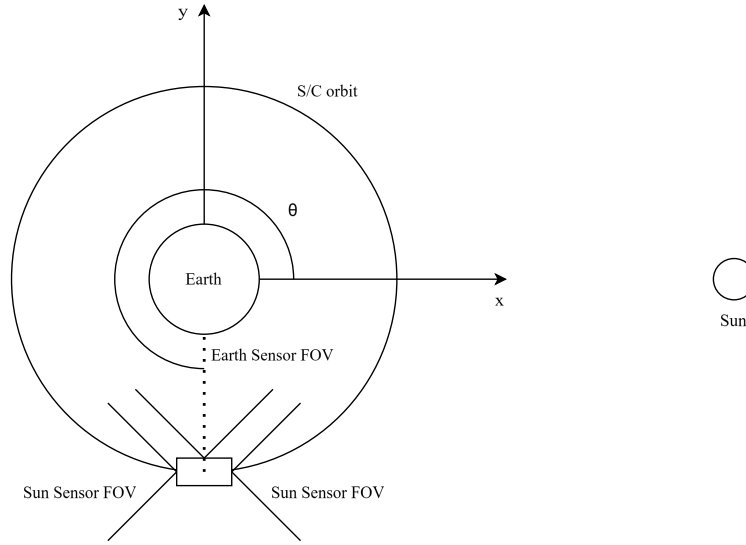


Figure 5: Initial True anomaly

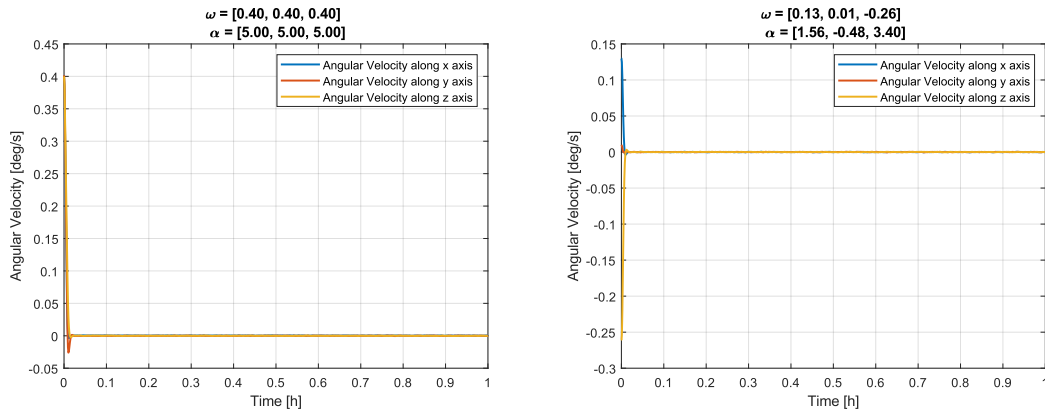
True anomaly $[\circ]$	$ \alpha $ $[\circ]$	$ \omega_x $ $[\circ/s]$	$ \omega_y $ $[\circ/s]$	$ \omega_z $ $[\circ/s]$
270	5	0.4	0.4	0.4

Table 7: Parameters at release [6].

making the spacecraft immediately spin, thus not leaving any interesting behaviour for comparison. The pointing is started with the last controlled attitude and velocities given from the slew manoeuvre.

4 Results

In this section results regarding de-tumbling, figure 6, carried out with randomized initial conditions and the related slew manoeuvres, figure 7, are reported. Finally, results showing performance comparison during the pointing phase with respect to the uncontrolled pointing phase are also displayed, figure 8. The plots on the left show the evolution with active control, while on the right the uncontrolled behavior is reported.



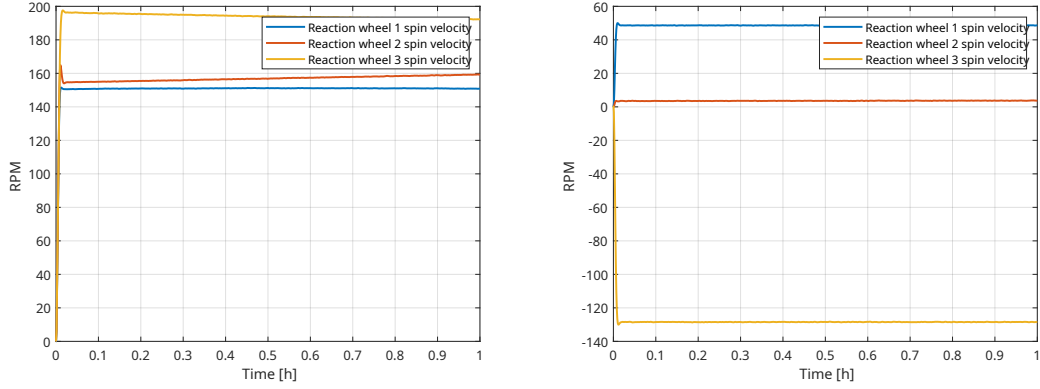


Figure 6: De-tumbling evolution from random initial angular velocities and deviation from the LVLH.

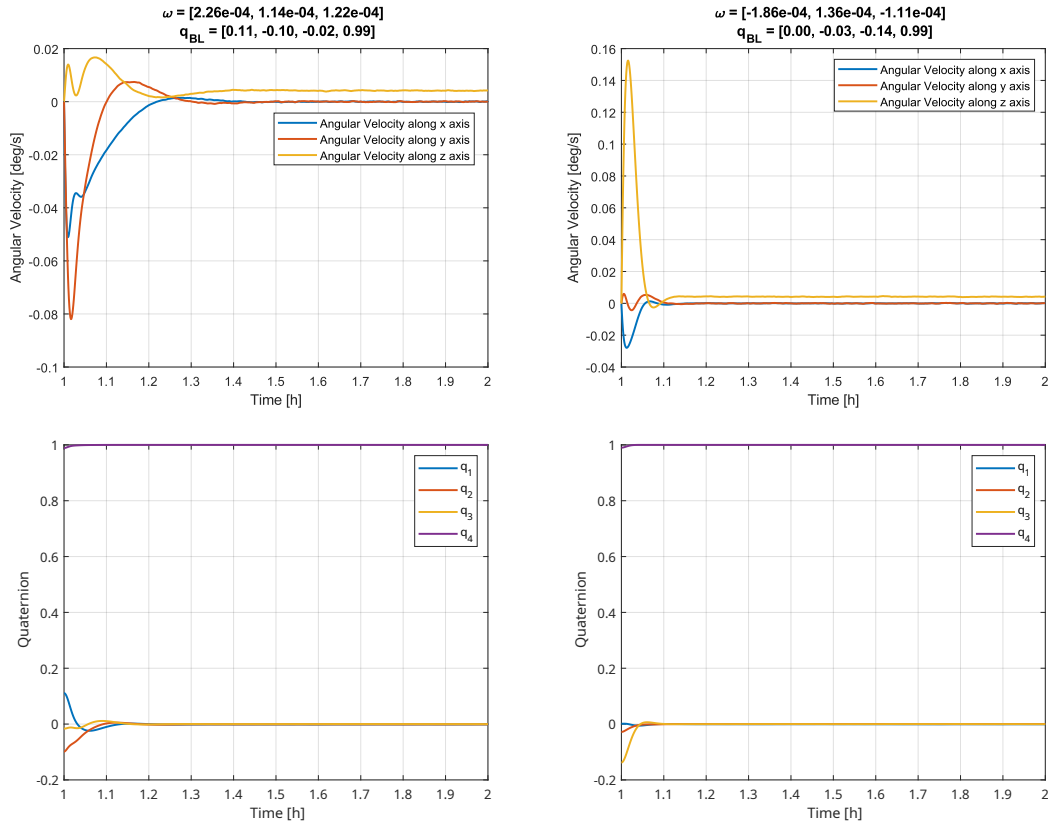
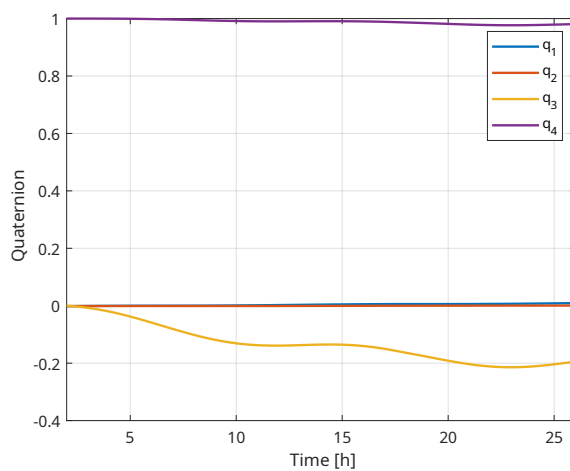
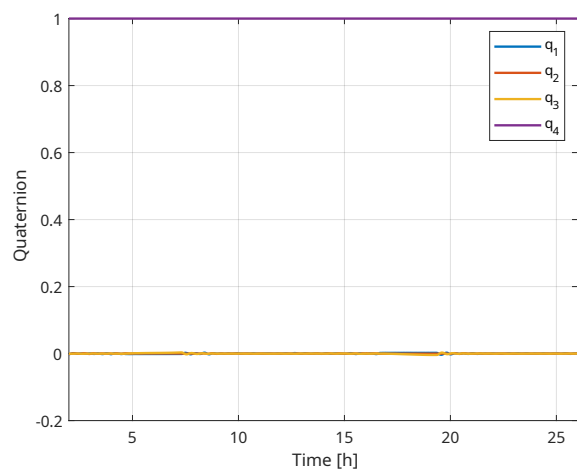
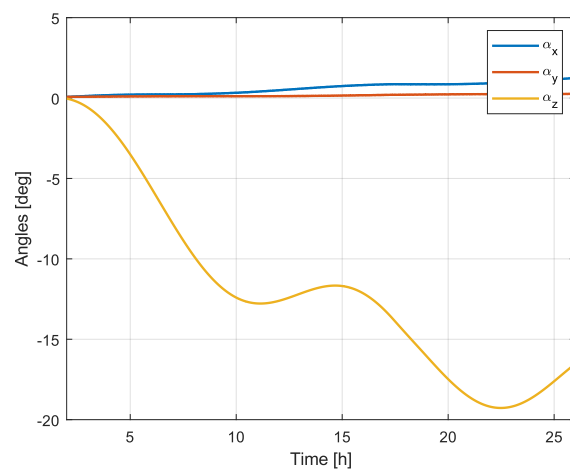
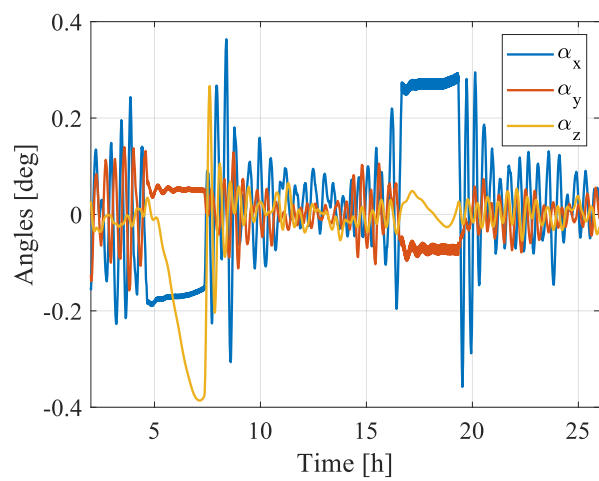


Figure 7: Slew manoeuvre evolution from de-tumbling final conditions.



Active control

No active control

Figure 8: Pointing performance.

5 Discussion

The simulation shows that the performance as outlined in section 1.1 are fulfilled when the spacecraft dynamics is subject to the control strategy employed. During the de-tumbling the proportional control is efficient in bringing the angular velocities below 10^{-5} rad/s in short time frames, around 10 minutes. Furthermore, the reaction wheels never saturate as predicted by the sizing.

The results of the evolution of the slew manoeuvre are shown in Figure 7. In different scenarios, it can be appreciated how the attitude error is brought to zero in terms of quaternions with angular velocities reaching steady state in a short period. The transition time from random attitude to nadir-pointing is achieved in around 10 minutes.

In the results for the pointing phase, figure 8, two critical areas along the orbit can be easily identified. These areas respectively correspond to the time when the spacecraft passes near true anomaly equal to zero, where the Sun sensors are lost; and when it passes behind the Earth and an eclipse takes place. During these periods, the prominent SRP torque deviates slightly the pointing, but thanks to the inertia wheel and active control the deviation is contained. Furthermore, when the Sun sensor blindness is ended, the pointing is immediately restored. During the whole phase, the requirement for the LV-LH of an angular deviation of $< 1^\circ$ is satisfied.

During the uncontrolled orbit coasting the performance of the satellite shows a gradually increasing error in the α_z angle. This behaviour is due to stabilized spin axis, which causes the angular velocities around the non-spinning axis to remain close to zero, but change the nominal rotating speed, causing a slowly increasing error in the pointing.

In conclusion, the results presented in section 5 show that the pointing requirements are fulfilled by the control strategy used. Furthermore, the results also show the performance achieved is a clear improvement in comparison with the uncontrolled case.

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